

MedGuardian AI



Executive Summary

Management of AI Products (MBACC ZG583)

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Problem

Patients managing three or more chronic prescriptions face two related risks that the healthcare system does not reliably catch today: missed doses, and drug combinations that cause harm. This is not a carelessness problem, rather, it is a visibility problem. A physician checks interactions against the medication list they know about, once, at the time of prescribing. A pharmacist checks again at pickup. Neither has any ongoing view of what a patient adds afterward: an over-the-counter painkiller, a supplement, a second prescription from a different specialist on a different record system. Missed doses and undetected interactions are a well-documented, preventable driver of hospital readmission in this population.

Target User / Customer

The primary end user is a patient aged roughly 45 to 75 managing three or more ongoing prescriptions for chronic conditions such as diabetes, hypertension, and high cholesterol. They need simple reminders and plain-language warnings, not alarm. The buyer who actually funds the product is different: health plans and pharmacy benefit managers (PBMs) bear the cost of the readmissions this product helps prevent, making them the realistic go-to-market channel, paying on a per-member-per-month basis for a measurable safety outcome.

Product Concept

MedGuardian AI is a medication adherence and drug-interaction copilot. A patient scans or searches for medication into the app; the system checks it against their current list using a real, FDA-grounded interaction reference; and any risk is flagged, with a mandatory pharmacist review sitting between the flag and anything the patient actually sees. Daily check-ins track adherence over time. The product is designed to be embedded inside an existing pharmacy, insurer, or hospital app, rather than compete as a new destination app.

Why AI?

The interaction-checking half of this product does not need a novel model, it runs on a deterministic, rules-based lookup against real pharmacological data (DrugBank, RxNorm, openFDA), which is exactly why it is buildable now rather than a multi-year research bet. AI earns its place in the second half of the job: turning a structured risk flag into something a worried patient can actually read and act on, in plain language, without sounding alarmist. A rules engine can tell a patient that two drugs interact; it cannot explain what that means for them in a way that builds trust instead of panic. That is a generation problem, not a lookup problem, and it is kept deliberately behind a human pharmacist for anything above low risk, so AI accelerates the explanation without ever making the safety call itself.

At a Glance

47 passing tests	$\geq 95\%$ over 60 days	$< 5\%$	Non-SaMD
Zero-AI safety core	Pharmacist agreement gate	False-positive ceiling	Regulatory posture

Recommendation

PILOT

The core rules-based engine, the mandatory pharmacist-review workflow, and a working Phase 1 prototype are already built and tested: 47 passing tests, real FDA drug data, zero live-AI dependency in the safety-critical path. The next step is not more prototyping; it is a scoped, time-boxed pilot with a real pharmacist partner, measured against the acceptance gates already defined for it: $\geq 95\%$ pharmacist agreement with the system’s flagged risk level, sustained over 60 days, and a false-positive rate under 5%. Scaling automation further, or making the generative explanation layer a default rather than a preview, should wait until that pilot data exists.